Table of contents

\\USER\Investigators\Michelmann\NeurWEM\AAHead_Scout_64ch-head-coil

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.2 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24

Resolution - Acceleration

Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
Phase	0.0 mm
Read	0.0 mm
Shift	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	260 mm
F >> H	260 mm
R >> L	205 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Dynamic

Dynamic Mode	Standard
Flip Angle	8 deg
Measurements	1
Time to Center	6.2 s

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Save Original Images	On
Contrasts	1
TE	1.37 ms
TR	3.2 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\t1_mprage_sag_p2_iso
TA: 5:21 min Coil Selection: Auto Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.9 mm
TR	2300.0 ms
TE	2.32 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	2300.0 ms
TE	2.32 ms
Magn. Preparation	Non-sel. IR
TI	900 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.9 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	GRAPPA

Resolution - Acceleration

Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.9 mm
TR	2300.0 ms
Multi-Slice Mode	Single Shot
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	Isocenter
Phase	0.0 mm
Read	0.0 mm
Shift	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P

System - Miscellaneous

Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	230 mm
F >> H	230 mm
R >> L	216 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	4.000

Physio - Signal

1st Signal/Mode	None
TR	2300.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	900 ms
Dark Blood	Off
FoV Read	230 mm
FoV Phase	100.0 %
Phase Resolution	100 %
Dynamic Mode	Standard

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	Non-sel. IR
Save Original Images	On
TE	2.32 ms

Inline - Cardiac

TR	2300.0 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.06 ms
Asymmetric Echo	Allowed
Turbo Factor	240

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\T2w_p2_s2_.9mm
TA: 3:07 min Coil Selection: Auto Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	240
Phase Oversampling	10 %
Slice Oversampling	0.0 %
FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.90 mm
TR	3200.0 ms
TE	564.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	3200.0 ms
TE	564.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.90 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	GRAPPA
Total Factor	4
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	240
Phase Oversampling	10 %
Slice Oversampling	0.0 %
FoV Read	230 mm
FoV Phase	100.0 %
Slice Thickness	0.90 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	Isocenter
Phase	0.0 mm
Read	0.0 mm
Shift	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	230 mm
F >> H	230 mm
R >> L	216 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FoV Read	230 mm
FoV Phase	100.0 %
Phase Resolution	100 %
Dynamic Mode	Standard

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	None

Inline - Cardiac

Save Original Images	On
TE	564.00 ms
TR	3200.0 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear
Bandwidth	751 Hz/Px
Echo Spacing	3.42 ms
Turbo Factor	314
Echo Train Duration	1070 ms

Sequence - Part 2

Introduction	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_AP
TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_PA

TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us

Sequence - Special

EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run1_1back
TA: 7:23 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	288
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	288
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run1_2back

TA: 6:11 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us

Sequence - Special

Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run2_1back

TA: 7:23 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	288
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	288
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run2_2back
TA: 6:11 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us

Sequence - Special

Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_AP

TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us

Sequence - Special

EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_PA
TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us

Sequence - Special

EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run3_1back

TA: 7:23 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	288
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	288
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us

Sequence - Special

Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run3_2back
TA: 6:11 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run4_1back

TA: 7:23 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	288
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	288
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_run4_2back

TA: 6:11 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P3.0 H6.0 mm
! Orientation	T > C-20.0
! Rotation	0.00 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us

Sequence - Special

Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_AP

TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\SE_DistortionMap_PA
TA: 27 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
TE	63.40 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	6831.0 ms
TE	63.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	6831.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6831.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2004 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
EPI noise scans	0
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Michelmann\NeurWEM\epi_MST

TA: 8:47 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	35.00 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	65 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	344
Delay in TR	0.00 ms

Resolution - Common

FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	208 mm
FoV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P3.0 H6.0
Phase	0.8 mm
Read	0.0 mm
Shift	6.7 mm
Initial Orientation	T > C
T > C	-20.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm

System - Adjustments

CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P3.0 H6.0 mm
Orientation	T > C-20.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.147532 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Active
Meas[2]	Active
Meas[3]	Ignore
Motion Correction	Off
Spatial Filter	Off
Measurements	344
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	4820 us

Sequence - Special

Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
EPI noise scans	0
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off